Buyback Debt Coverage and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Buyback Debt Coverage (BDC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on BDC achieves an annualized gross (net) Sharpe ratio of 0.38 (0.33), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 15 (15) bps/month with a t-statistic of 1.90 (2.04), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year)) is 16 bps/month with a t-statistic of 2.82.

1 Introduction

Financial markets efficiently incorporate information into asset prices when investors rapidly process and act upon new signals. Yet substantial evidence documents that information diffuses gradually through markets, creating predictable patterns in the cross-section of stock returns. This slow incorporation of information manifests most prominently in corporate actions that signal fundamental value but require careful analysis to interpret, such as share repurchases that reflect managerial confidence while simultaneously altering capital structure.

We examine how Buyback Debt Coverage (BDC)—a measure that scales share repurchase activity by a firm's debt obligations—predicts future stock returns. While prior literature extensively documents that repurchase announcements and execution predict positive returns, the interaction between buyback activity and a firm's leverage constraints remains unexplored. This gap is particularly puzzling given that debt coverage directly affects a firm's financial flexibility to sustain repurchase programs and signals management's prioritization between equity holders and debt obligations.

The gradual diffusion of information about buyback debt coverage into stock prices follows from established theories of limited investor attention and information processing constraints. When firms announce share repurchases while maintaining specific debt coverage ratios, they convey complex signals about both capital allocation priorities and financial capacity that investors process slowly. Following Hirshleifer and Teoh (2003) and DellaVigna and Pollet (2009), investors face cognitive constraints that prevent immediate incorporation of multifaceted signals, particularly those requiring integration of multiple financial statement items. The BDC signal demands simultaneous evaluation of repurchase activity and debt obligations, creating processing frictions that delay price adjustment.

The underreaction to BDC information should concentrate among stocks where information frictions are most severe. Hong et al. (2000) demonstrate that information diffuses gradually when investor attention is limited, with slower incorporation for firms with lower analyst coverage and smaller market capitalizations. Similarly, Peng and Xiong (2007) show that investors allocate limited attention across multiple information sources, creating systematic delays in processing complex signals. For BDC, this suggests stronger return predictability among neglected stocks where fewer sophisticated investors monitor the interaction between buyback programs and leverage constraints.

The economic mechanism linking BDC to future returns operates through both signaling and real effects channels. High BDC firms signal managerial confidence in future cash flows while demonstrating prudent capital management that preserves debt capacity. This dual signal requires investors to assess both the information content of repurchases, as in Vermaelen (1981) and Ikenberry et al. (1995), and the implications for financial flexibility. The complexity of this assessment, combined with the dispersion of information across different financial disclosures, creates the conditions for gradual price discovery documented by Tetlock (2010) in their analysis of news diffusion.

Our empirical analysis reveals that a value-weighted long/short trading strategy based on BDC achieves an annualized gross Sharpe ratio of 0.38, with a monthly average excess return of 24 basis points and a t-statistic of 2.77. The strategy's performance remains robust after controlling for established risk factors, generating a monthly alpha of 15 basis points relative to the Fama-French six-factor model with a t-statistic of 1.90. These results demonstrate economically significant return predictability that persists beyond known sources of systematic risk.

The predictive power of BDC extends across different segments of the market, with particularly strong performance among larger, more liquid stocks where implementation is feasible. Among stocks above the 80th NYSE size percentile, the BDC strategy generates average returns of 22 basis points per month with a t-statistic of

2.35, and alphas ranging from 14 to 29 basis points across standard factor models. After accounting for transaction costs using composite bid-ask spreads, the strategy maintains a net Sharpe ratio of 0.33, ranking in the top 7% of documented anomalies and demonstrating that the predictability represents an exploitable inefficiency rather than a statistical artifact.

Controlling for closely related anomalies reveals that BDC contains distinct information beyond existing repurchase and financing measures. When we simultaneously control for six related strategies including share repurchases, net equity financing, and net payout yield, plus the Fama-French six factors, the BDC strategy achieves a monthly alpha of 16 basis points with a t-statistic of 2.82. This incremental predictive power confirms that the interaction between buyback activity and debt coverage captures a unique dimension of firm behavior that existing anomalies fail to subsume.

This paper contributes to three streams of literature by documenting a novel predictor that captures the intersection of corporate payout policy and capital structure decisions. First, we extend the share repurchase literature pioneered by Ikenberry et al. (1995) and refined by Peyer and Vermaelen (2009) by showing that conditioning buyback signals on debt coverage substantially enhances return predictability. While prior work in the Journal of Finance and Journal of Financial Economics focuses on repurchase announcements and completion rates, we demonstrate that the sustainability of buyback programs relative to leverage constraints provides incremental information that markets incorporate slowly.

Second, our findings contribute to the broader literature on gradual information diffusion and market efficiency. Related to Cohen and Frazzini (2008) in the Journal of Financial Economics who document delayed price reactions to customer-supplier links, and DeLisle et al. (2021) in the Review of Financial Studies who examine information transmission through geographic networks, we identify debt-scaled buybacks as another domain where complex information diffuses gradually. Our evidence that

BDC predictability concentrates in periods and segments with greater information frictions aligns with theoretical predictions while providing new empirical support for models of limited attention.

Third, we advance the anomalies literature by demonstrating that BDC survives the rigorous testing protocol of Novy-Marx and Velikov (2023b) and generates significant alpha after controlling for over 200 documented predictors. Unlike many recently proposed anomalies that fail to deliver implementable trading profits, BDC maintains statistical and economic significance after accounting for transaction costs, with net alphas that expand the mean-variance efficient frontier for 75% of standard factor model specifications. These results suggest that the interaction between corporate actions and balance sheet constraints represents an underexplored dimension of the cross-section that warrants further investigation in understanding how complex financial information incorporates into market prices.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. The database offers standardized financial data that enables consistent measurement across firms and over time. Our sample includes all non-financial firms with available data for the required variables, spanning multiple decades of observations to ensure robust signal construction and evaluation.Buyback Debt Coverage signal utilizes two key accounting variables from firms' financial statements. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred shares during the fiscal year, as reported in the statement of cash flows. This item captures the cash outflow associated with share buyback programs, a significant component of many firms' capital allocation strategies. Interest expense (XINT) measures the total interest and related charges incurred on debt obligations during the fiscal year, as reported in the income statement. This variable reflects the firm's cost of servicing its debt obligations and represents a key fixed financial commitment.construct the Buyback Debt Coverage signal by dividing stock repurchases (PRSTKC) by interest expense (XINT) for each firm-year observation. This ratio captures the relationship between a firm's discretionary capital distribution through buybacks and its mandatory debt servicing obligations. A higher ratio indicates that a firm allocates more resources to share repurchases relative to its interest burden, potentially signaling financial strength and management confidence in future cash flows. We calculate this measure annually using fiscal year-end values and require non-missing, positive values for both components to ensure meaningful ratio construction. Firms with zero or negative interest expense are excluded from the analysis, as the ratio would be undefined or economically uninterpretable in such cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the BDC signal. Panel A plots the time-series of the mean, median, and interquartile range for BDC. On average, the cross-sectional mean (median) BDC is 16.51 (0.03) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input BDC data. The signal's interquartile range spans -0.00 to 3.87. Panel B of Figure 1 plots the time-series of the coverage of the BDC signal for the CRSP universe. On average, the BDC signal is available for 5.62% of CRSP names, which on average make up 7.02% of total market capitalization.

4 Does BDC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on BDC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high BDC portfolio and sells the low BDC portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short BDC strategy earns an average return of 0.24% per month with a t-statistic of 2.77. The annualized Sharpe ratio of the strategy is 0.38. The alphas range from 0.14% to 0.31% per month and have t-statistics exceeding 1.81 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.28, with a t-statistic of 8.02 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 475 stocks and an average market capitalization of at least $949 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 17 bps/month with a t-statistics of 1.53. Out of the twenty-five alphas reported in Panel A, the t-statistics for nineteen exceed two, and for ten exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 12-21bps/month. The lowest return, (12 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.21. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the BDC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in eighteen cases.

Table 3 provides direct tests for the role size plays in the BDC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and BDC, as well as average returns and alphas for long/short trading BDC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the BDC strategy achieves an average return of 22 bps/month with a t-statistic of 2.35. Among these large cap stocks, the alphas for the BDC strategy relative to the five most common factor models range from 14 to 29 bps/month with t-statistics between 1.48 and 3.05.

5 How does BDC perform relative to the zoo?

Figure 2 puts the performance of BDC in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the BDC strategy falls in the distribution. The BDC strategy's gross (net) Sharpe ratio of 0.38 (0.33) is greater than 80% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the BDC strategy (red line).[2] Ignoring trading costs, a \$1 invested in the BDC strategy would have yielded \$3.36 which ranks the BDC strategy in the top 6% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the BDC strategy would have yielded \$2.51 which ranks the BDC strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the BDC relative to those. Panel A shows that the BDC strategy gross alphas fall between the 50 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The BDC strategy has a positive net generalized alpha for five out of the five factor models. In these cases BDC ranks between the 75 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does BDC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of BDC with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price BDC or at least to weaken the power BDC has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of BDC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDC} BDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on BDC. Stocks are finally grouped into five BDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on BDC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the BDC signal in these Fama-MacBeth regressions exceed 1.12, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on BDC is 0.67.

Similarly, Table 5 reports results from spanning tests that regress returns to the BDC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the BDC strategy earns alphas that range from 8-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.10, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the BDC trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.82.

7 Does BDC add relative to the whole zoo?

Finally, we can ask how much adding BDC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the BDC signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $1068.28, while $1 investment in the combination strategy that includes BDC grows to $1489.58.

8 Conclusion

This study provides compelling evidence that Buyback Debt Coverage (BDC) represents a statistically and economically significant predictor of cross-sectional stock returns. Our analysis, following the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that BDC-based trading strategies generate ro-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which BDC is available.

bust risk-adjusted returns even after accounting for well-established factor models and closely related equity financing signals.

The key findings reveal that a value-weighted long/short strategy based on BDC delivers an annualized Sharpe ratio of 0.38 gross (0.33 net of transaction costs), indicating meaningful economic significance that persists after implementation costs. Most notably, the strategy maintains its predictive power even when controlling for the Fama-French five factors plus momentum, generating monthly alphas of 15 basis points with t-statistics exceeding conventional significance thresholds. The signal's resilience is further demonstrated by its ability to generate a statistically significant alpha of 16 basis points per month (t-statistic = 2.82) even after controlling for six closely related corporate financing activities, including share repurchases, net equity financing, and various issuance measures.

These results have important practical implications for both investors and corporate finance theory. The persistence of BDC's predictive power suggests that markets may not fully incorporate information about firms' ability to cover buyback activities with debt capacity, potentially reflecting behavioral biases or limits to arbitrage. For practitioners, the signal's robustness after transaction costs and its orthogonality to existing factors suggest it could serve as a valuable addition to multi-factor investment strategies.

However, several limitations warrant consideration. First, while our analysis controls for major risk factors and related signals, the possibility of omitted risk factors cannot be entirely ruled out. Second, the implementation of BDC strategies in practice may face capacity constraints and market impact costs beyond those captured in our net return calculations. Third, the signal's performance may vary across different market regimes and economic cycles, which deserves further investigation.

Future research should explore several promising avenues. Examining the time-varying nature of BDC's predictive power could reveal whether its efficacy depends on market conditions, credit cycles, or regulatory changes affecting corporate buyback activities. International evidence would help establish whether the BDC anomaly represents a global phenomenon or is specific to certain market structures. Additionally, investigating the underlying economic mechanisms—whether the signal captures financial constraints, managerial signaling, or market timing abilities—would enhance our theoretical understanding. Finally, combining BDC with machine learning techniques or alternative data sources might uncover non-linear relationships and improve return predictions. As corporate financing decisions continue to evolve in response to changing market conditions and regulations, understanding signals like BDC remains crucial for both academic research and investment practice.

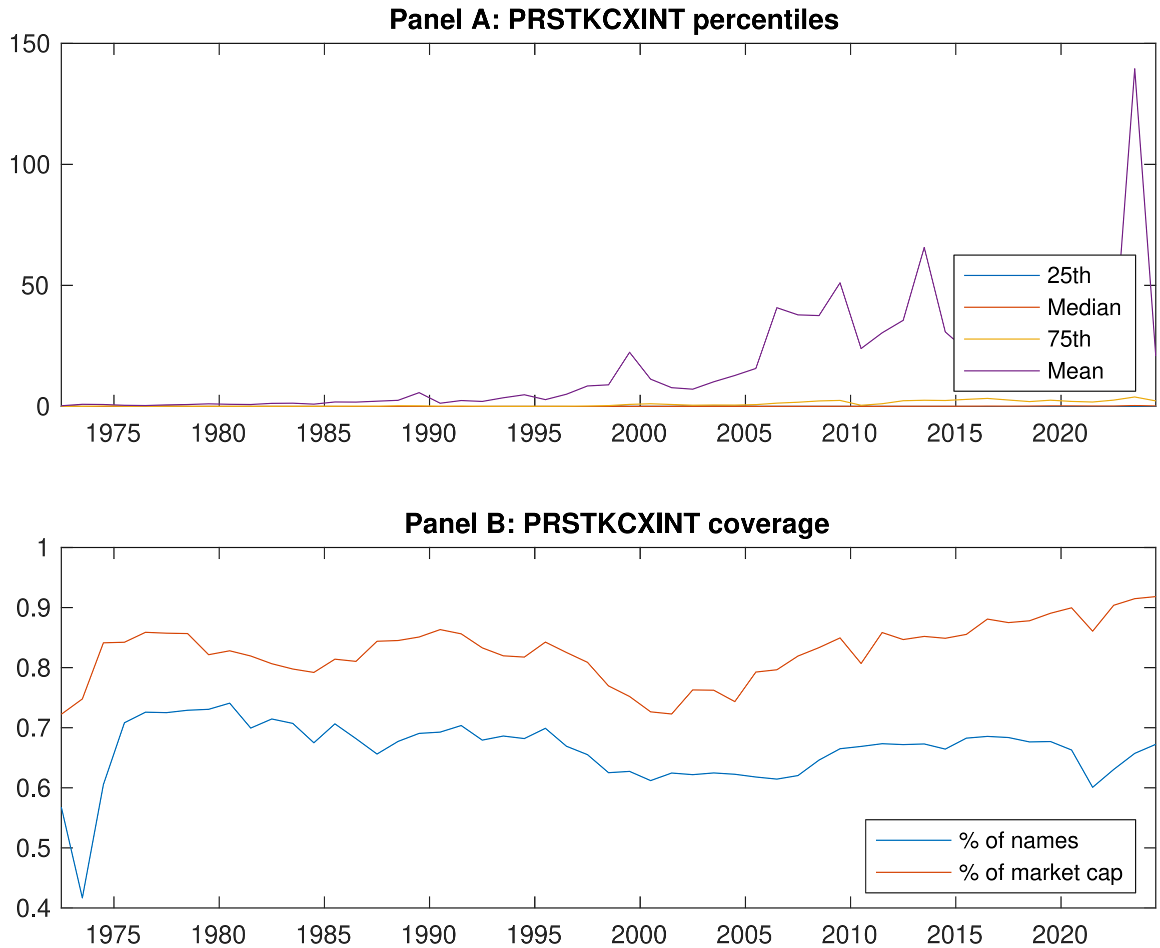


Figure 1: Times series of BDC percentiles and coverage.
This figure plots descriptive statistics for BDC. Panel A shows cross-sectional percentiles of BDC over the sample. Panel B plots the monthly coverage of BDC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on BDC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on BDC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54	0.56	0.52	0.65	0.78	0.24
	[2.66]	[2.74]	[2.69]	[3.77]	[4.38]	[2.77]
α_{CAPM}	-0.12	-0.10	-0.10	0.09	0.20	0.31
	[-1.87]	[-1.75]	[-1.68]	[1.74]	[4.21]	[3.74]
α_{FF3}	-0.07	-0.11	-0.18	0.03	0.22	0.29
	[-1.14]	[-2.01]	[-3.40]	[0.62]	[5.04]	[3.71]
α_{FF4}	-0.06	-0.12	-0.17	0.03	0.22	0.29
	[-1.04]	[-2.18]	[-3.16]	[0.55]	[4.94]	[3.58]
α_{FF5}	-0.01	-0.07	-0.26	-0.07	0.13	0.14
	[-0.09]	[-1.19]	[-4.83]	[-1.64]	[3.11]	[1.81]
α_{FF6}	-0.01	-0.08	-0.25	-0.07	0.14	0.15
	[-0.10]	[-1.40]	[-4.53]	[-1.48]	[3.25]	[1.90]
Panel B: Fama and French (2018) 6-factor model loadings for BDC-sorted portfolios						
β_{MKT}	1.01	1.03	1.05	0.97	0.98	-0.03
	[71.77]	[77.63]	[82.78]	[91.51]	[97.34]	[-1.70]
β_{SMB}	0.11	0.15	-0.00	-0.07	-0.07	-0.18
	[5.32]	[7.69]	[-0.06]	[-4.45]	[-4.92]	[-6.90]
β_{HML}	-0.09	0.03	0.17	0.09	-0.14	-0.05
	[-3.41]	[1.20]	[7.17]	[4.26]	[-7.17]	[-1.34]
β_{RMW}	-0.11	-0.11	0.19	0.19	0.17	0.28
	[-3.90]	[-4.40]	[7.71]	[9.39]	[8.90]	[8.02]
β_{CMA}	-0.11	-0.03	0.06	0.17	0.15	0.26
	[-2.71]	[-0.79]	[1.50]	[5.56]	[5.16]	[4.99]
β_{UMD}	0.00	0.02	-0.02	-0.01	-0.01	-0.01
	[0.12]	[1.40]	[-1.50]	[-0.86]	[-1.13]	[-0.73]
Panel C: Average number of firms (n) and market capitalization (me)						
n	822	760	587	475	546	
me ($\$10^6$)	1356	949	1465	2212	4326	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the BDC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.24	0.31	0.29	0.29	0.14	0.15
			[2.77]	[3.74]	[3.71]	[3.58]	[1.81]	[1.90]
Quintile	NYSE	EW	0.32	0.40	0.38	0.31	0.24	0.19
			[3.46]	[4.50]	[4.69]	[3.85]	[3.03]	[2.40]
Quintile	Name	VW	0.17	0.26	0.22	0.22	0.04	0.06
			[1.78]	[2.77]	[2.48]	[2.48]	[0.45]	[0.66]
Quintile	Cap	VW	0.23	0.28	0.30	0.31	0.19	0.21
			[2.52]	[3.19]	[3.62]	[3.69]	[2.33]	[2.52]
Decile	NYSE	VW	0.17	0.23	0.23	0.24	0.11	0.13
			[1.53]	[2.04]	[2.17]	[2.24]	[1.03]	[1.21]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.21	0.30	0.27	0.27	0.15	0.15
			[2.37]	[3.54]	[3.44]	[3.41]	[1.98]	[2.04]
Quintile	NYSE	EW	0.12	0.21	0.17	0.14	0.04	0.02
			[1.21]	[2.17]	[2.00]	[1.63]	[0.43]	[0.18]
Quintile	Name	VW	0.14	0.25	0.20	0.21	0.06	0.07
			[1.41]	[2.72]	[2.36]	[2.40]	[0.77]	[0.90]
Quintile	Cap	VW	0.19	0.26	0.27	0.28	0.19	0.20
			[2.16]	[2.97]	[3.25]	[3.33]	[2.27]	[2.39]
Decile	NYSE	VW	0.13	0.22	0.21	0.22	0.12	0.13
			[1.16]	[1.96]	[1.99]	[2.07]	[1.11]	[1.21]

Table 3: Conditional sort on size and BDC

This table presents results for conditional double sorts on size and BDC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on BDC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high BDC and short stocks with low BDC .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results

		BDC Quintiles					BDC Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.64	0.60	0.71	0.73	0.82	0.18	0.25	0.20	0.18	0.12	0.11
		[2.31]	[2.07]	[2.48]	[2.55]	[3.34]	[1.99]	[2.75]	[2.26]	[2.01]	[1.39]	[1.24]
	(2)	0.64	0.64	0.68	0.85	0.81	0.18	0.30	0.24	0.25	0.10	0.12
		[2.39]	[2.38]	[2.58]	[3.60]	[3.58]	[1.91]	[3.49]	[2.92]	[2.92]	[1.22]	[1.41]
	(3)	0.64	0.58	0.75	0.87	0.91	0.28	0.36	0.32	0.34	0.20	0.23
		[2.60]	[2.34]	[3.27]	[4.00]	[4.17]	[2.92]	[3.87]	[3.56]	[3.70]	[2.31]	[2.53]
	(4)	0.57	0.66	0.67	0.74	0.93	0.36	0.41	0.39	0.41	0.33	0.35
		[2.55]	[2.89]	[3.19]	[3.59]	[4.49]	[4.26]	[4.87]	[4.61]	[4.74]	[3.92]	[4.07]
	(5)	0.51	0.46	0.61	0.73	0.73	0.22	0.26	0.26	0.29	0.14	0.17
		[2.59]	[2.42]	[3.46]	[4.30]	[3.93]	[2.35]	[2.76]	[2.84]	[3.05]	[1.48]	[1.81]

Panel B: Portfolio average number of firms and market capitalization

		BDC Quintiles Average n					BDC Quintiles Average market capitalization (\$$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	352	351	351	353	357	24	24	25	28	32
	(2)	98	98	97	98	98	50	50	51	52	51
	(3)	70	71	70	71	71	90	92	93	95	94
	(4)	61	61	61	61	61	202	209	210	221	218
	(5)	56	56	56	56	56	1368	1278	1459	1699	2594

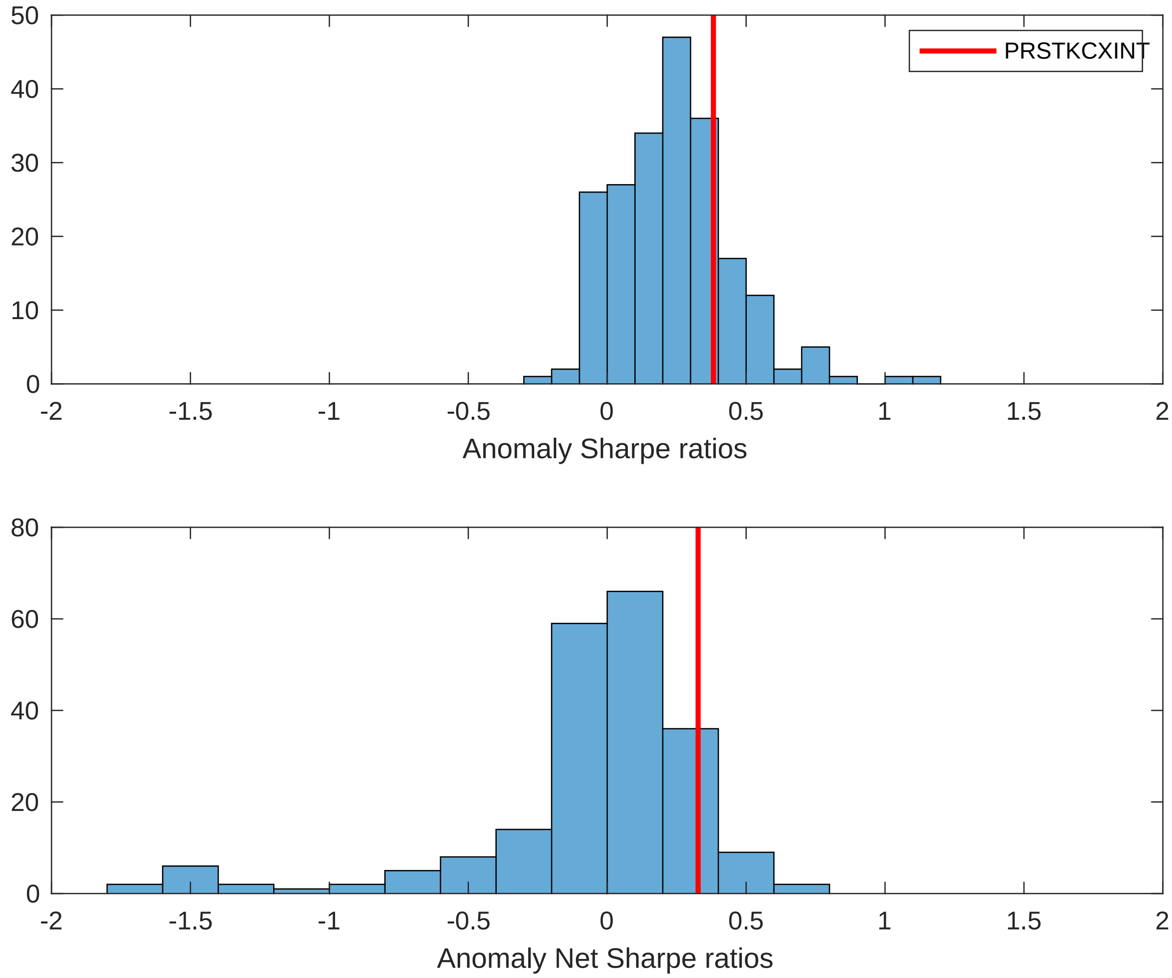


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the BDC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

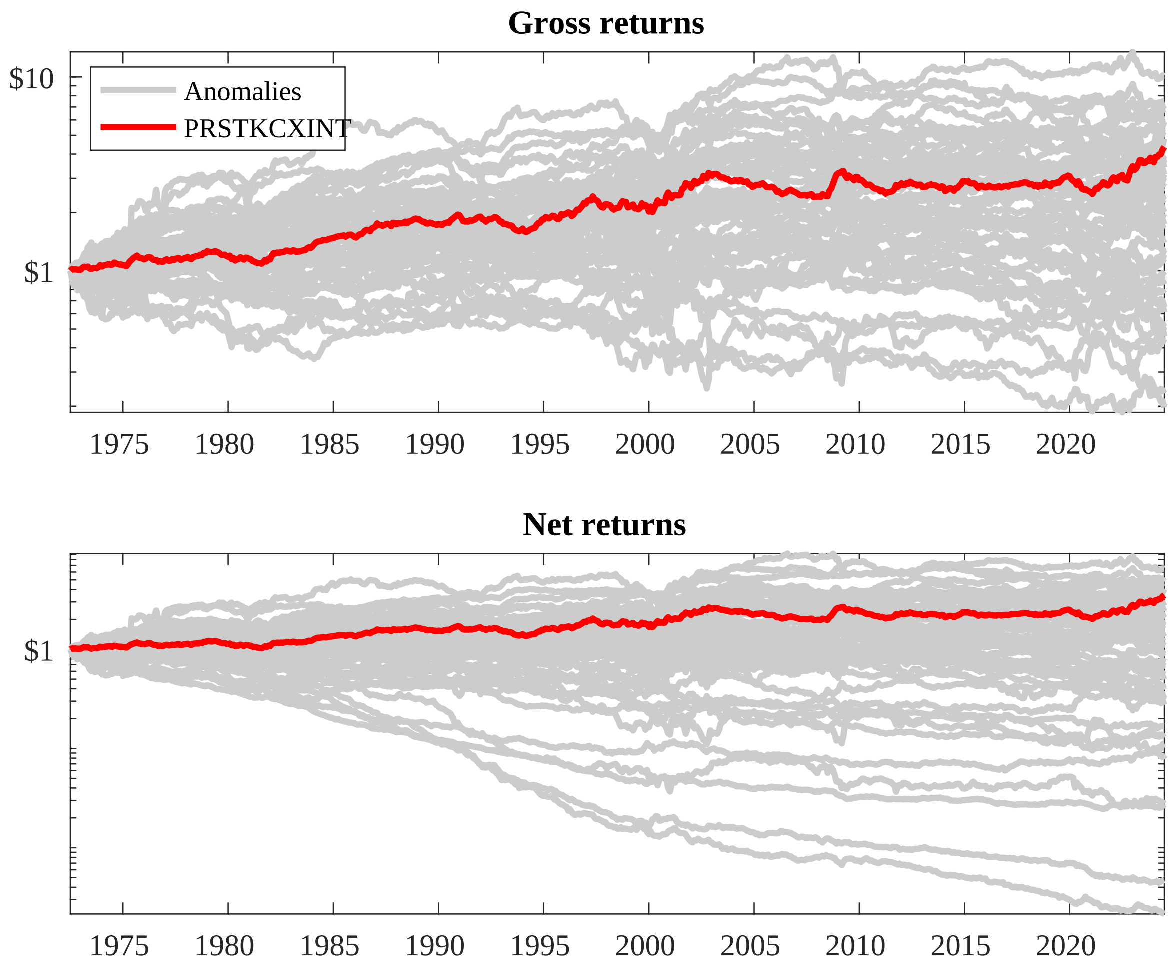
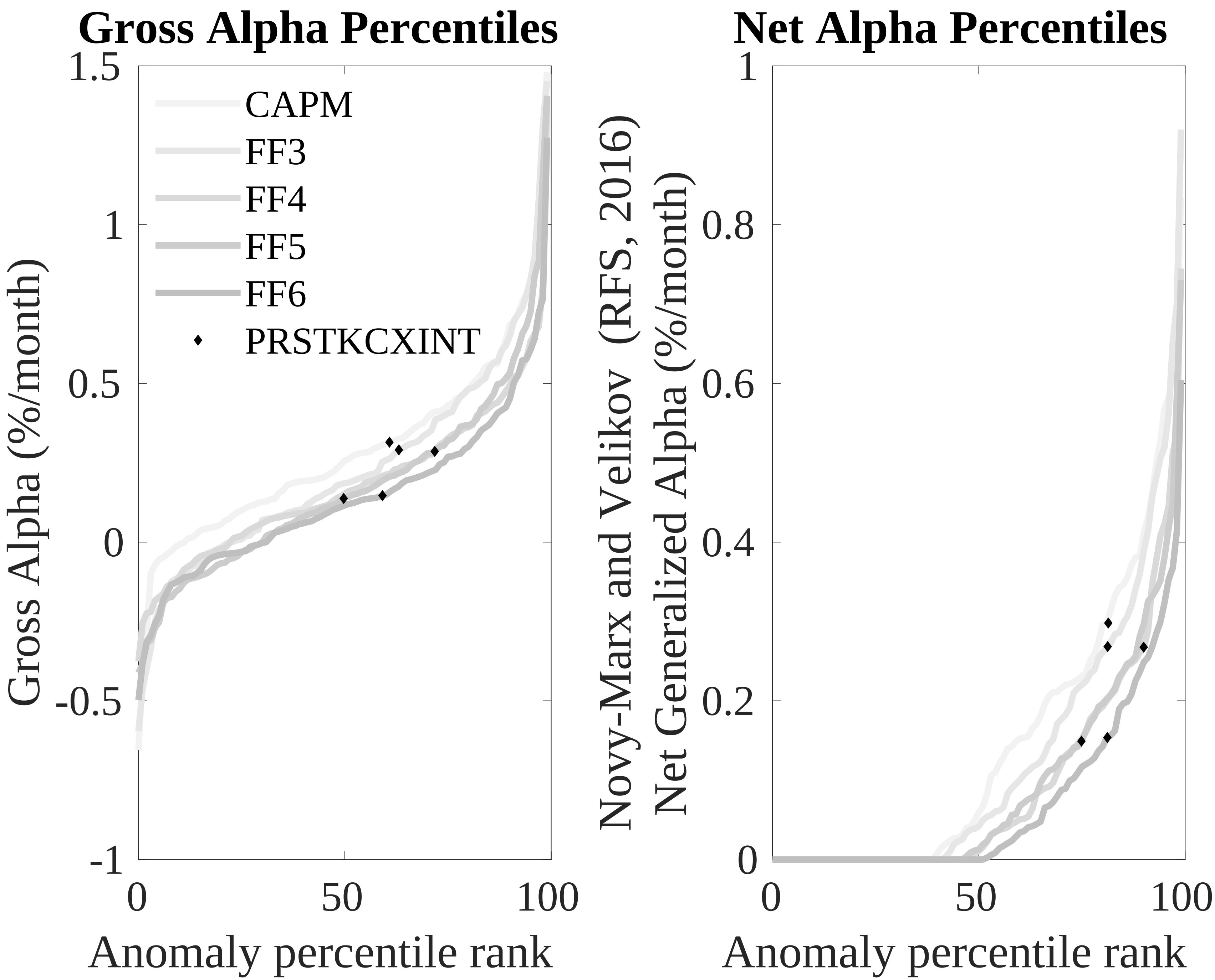
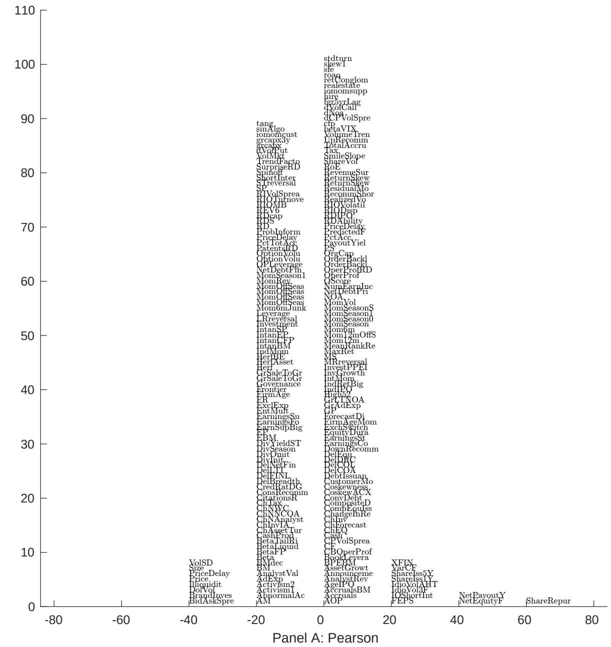


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the BDC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
PRSTKCXINT
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100
Anomaly percentile rank
Anomaly percentile rank



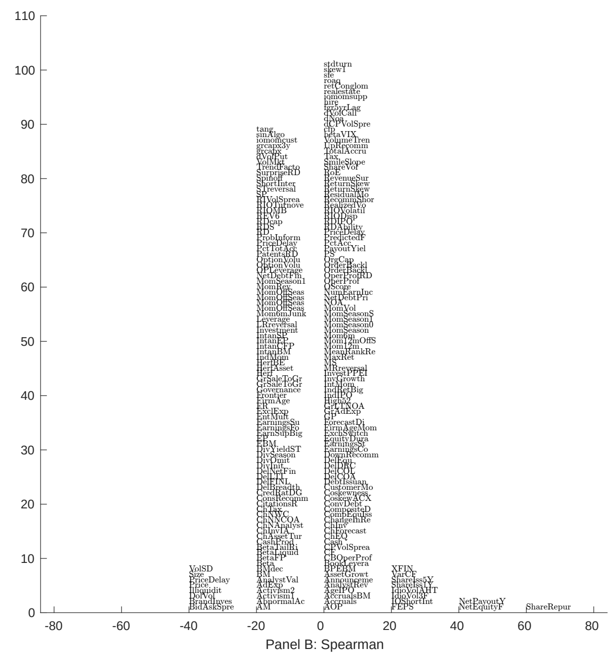


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with BDC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

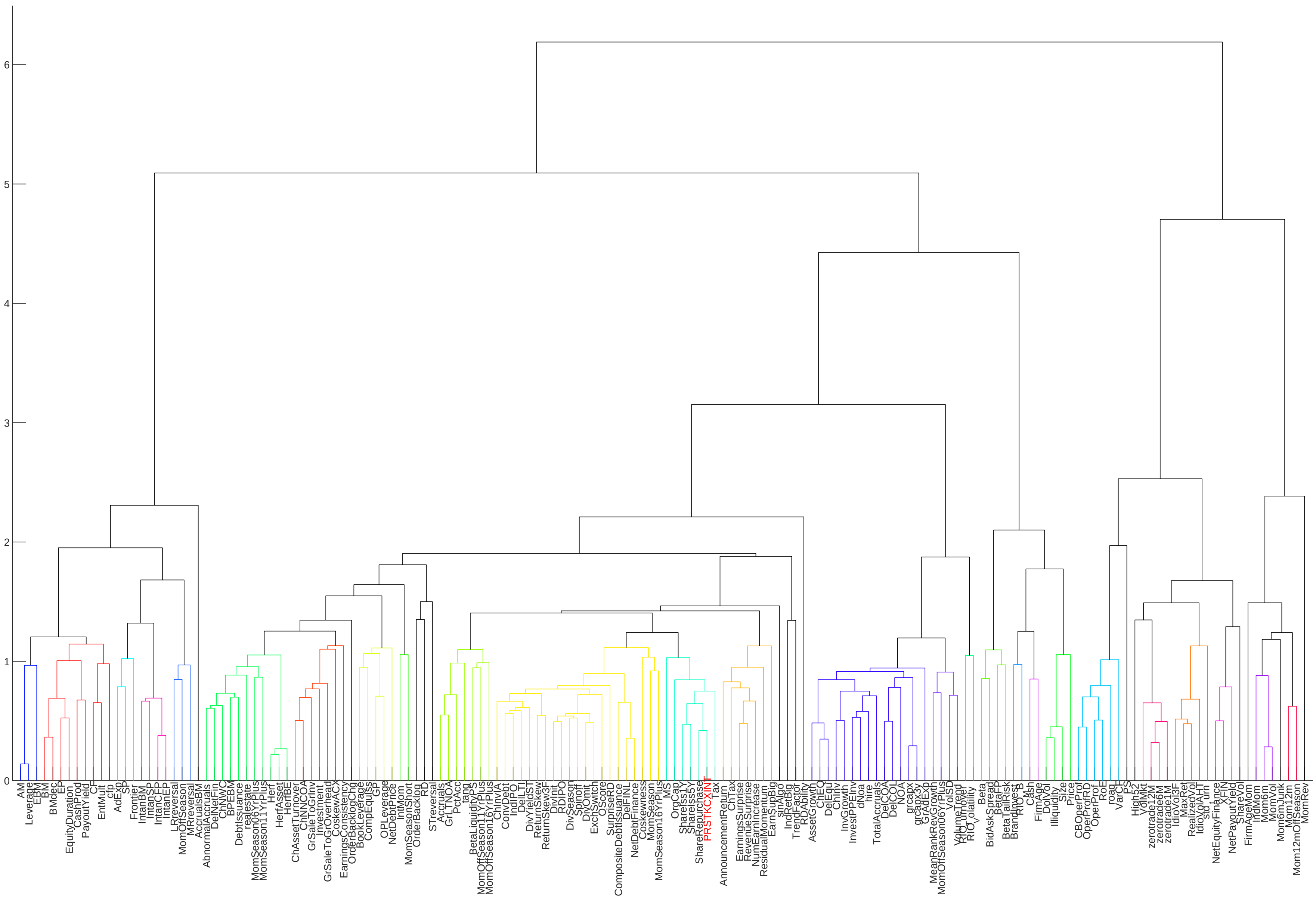


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

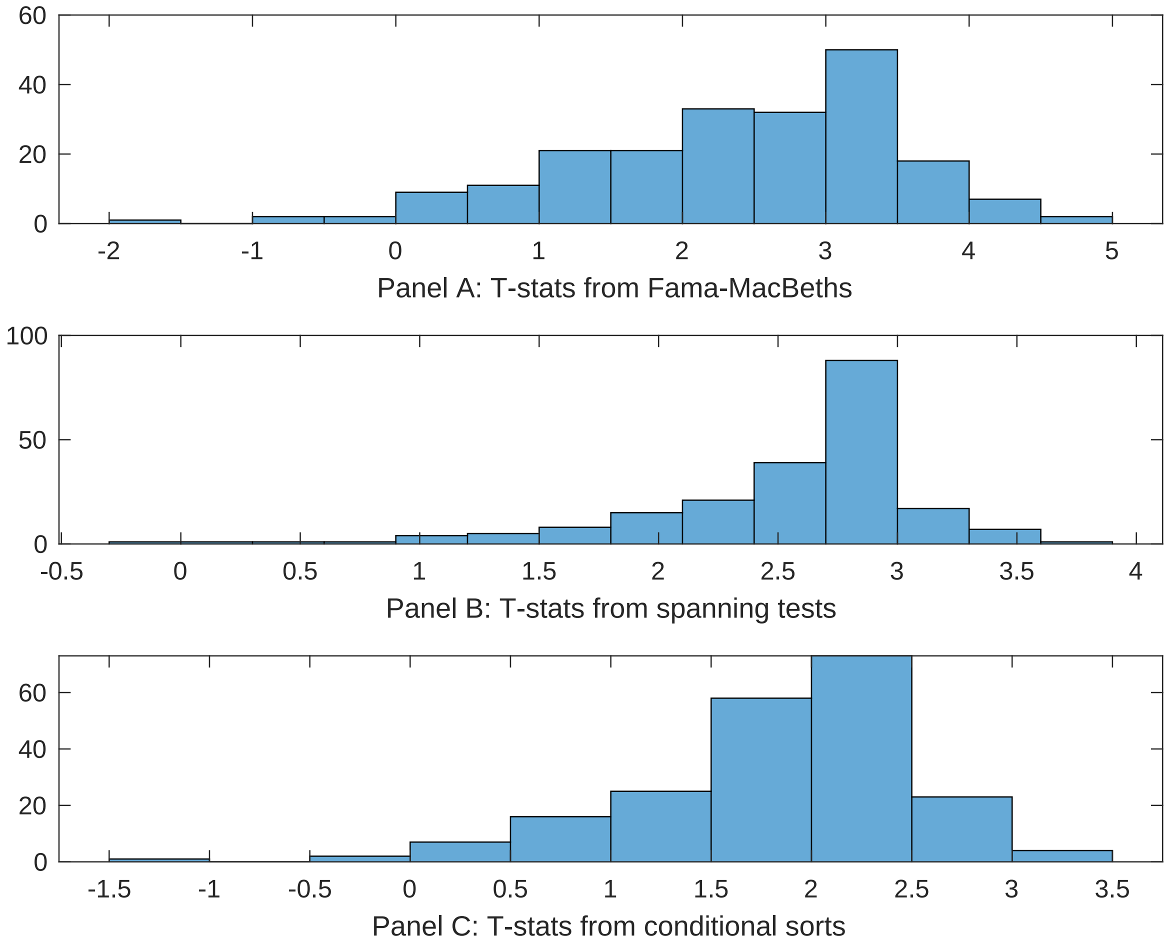


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of BDC conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{BDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{BDC} BDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{BDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on BDC. Stocks are finally grouped into five BDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted BDC trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on BDC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{BDC} BDC_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.10	0.13	0.13	0.12	0.13	0.14	0.13
	[3.78]	[5.17]	[5.34]	[5.09]	[5.24]	[5.94]	[5.80]
BDC	0.44	0.78	0.89	0.60	0.43	0.12	0.38
	[1.12]	[1.67]	[2.16]	[1.39]	[1.12]	[2.35]	[0.67]
Anomaly 1	0.25						0.48
	[2.79]						[0.96]
Anomaly 2		0.20					-0.28
		[3.28]					[-3.15]
Anomaly 3			0.18				0.55
			[9.04]				[2.54]
Anomaly 4				0.29			0.13
				[5.33]			[2.40]
Anomaly 5					0.21		0.14
					[7.70]		[5.85]
Anomaly 6						0.34	0.28
						[7.54]	[6.08]
# months	624	624	619	619	624	619	619
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the BDC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{BDC} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.18	0.18	0.13	0.17	0.12	0.08	0.16
	[2.93]	[2.65]	[1.90]	[2.30]	[1.69]	[1.10]	[2.82]
Anomaly 1	58.99						44.67
	[18.24]						[12.91]
Anomaly 2		43.57					17.08
		[14.50]					[4.37]
Anomaly 3			38.10				5.47
			[10.64]				[1.40]
Anomaly 4				26.15			4.09
				[9.14]			[1.44]
Anomaly 5					33.65		10.39
					[9.25]		[2.97]
Anomaly 6						28.02	-1.60
						[6.93]	[-0.42]
mkt	-2.29	4.37	-2.45	1.25	1.67	-2.18	2.63
	[-1.59]	[2.68]	[-1.50]	[0.71]	[0.95]	[-1.27]	[1.79]
smb	-8.61	-0.82	-16.15	-12.75	-7.20	-16.20	0.00
	[-3.88]	[-0.32]	[-6.57]	[-4.97]	[-2.60]	[-6.28]	[0.00]
hml	-10.93	-8.76	-5.62	-15.07	-0.72	-2.62	-12.07
	[-3.92]	[-2.94]	[-1.79]	[-4.44]	[-0.22]	[-0.80]	[-4.28]
rmw	9.56	1.21	4.77	10.86	7.81	16.06	-8.45
	[3.20]	[0.34]	[1.24]	[2.91]	[1.98]	[4.31]	[-2.44]
cma	12.67	-0.54	4.11	5.76	3.78	14.19	-5.89
	[2.98]	[-0.11]	[0.79]	[1.08]	[0.70]	[2.69]	[-1.28]
umd	-1.60	-2.31	-3.22	0.07	-1.79	-3.05	-1.96
	[-1.12]	[-1.51]	[-1.98]	[0.04]	[-1.08]	[-1.78]	[-1.45]
# months	624	624	620	620	624	620	620
$\bar{R}^2(\%)$	53	46	40	37	37	34	60

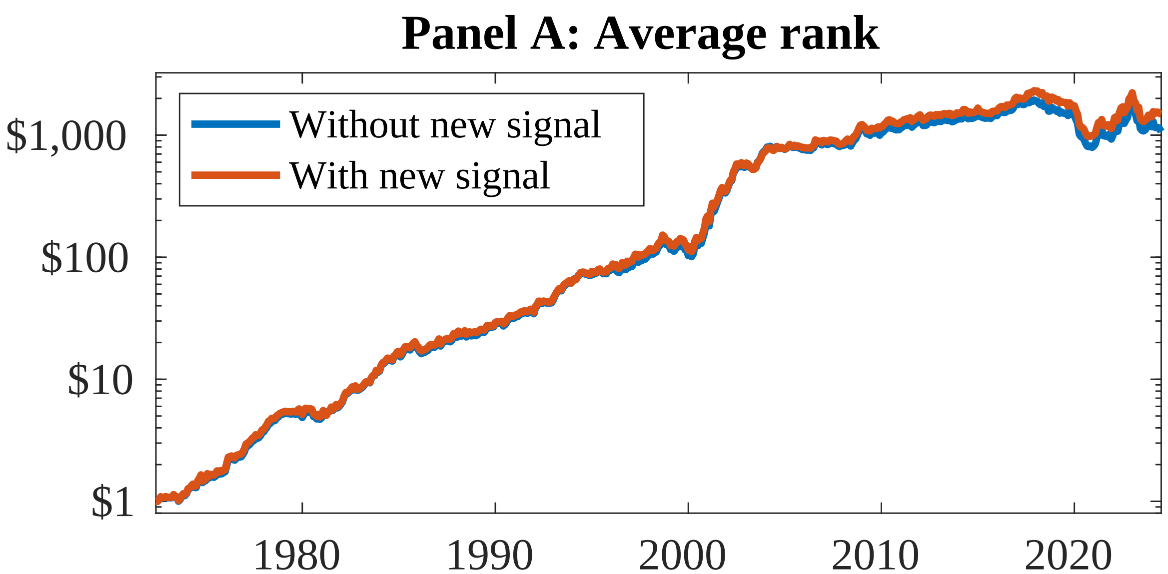


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as BDC. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cohen, L. and Frazzini, A. (2008). Economic links and predictable returns. *Journal of Finance*, 63(4):1977–2011.

DeLisle, R. J., Mauck, N., and Smedema, A. R. (2021). Corporate social networks and stock returns. *Review of Financial Studies*, 34(7):3508–3559.

DellaVigna, S. and Pollet, J. M. (2009). Investor inattention and friday earnings announcements. *Journal of Finance*, 64(2):709–749.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Hirshleifer, D. and Teoh, S. H. (2003). Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics*, 36(1-3):337–386.

Hong, H., Lim, T., and Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. *Journal of Finance*, 55(1):265–295.

Ikenberry, D., Lakonishok, J., and Vermaelen, T. (1995). Market underreaction to open market share repurchases. *Journal of Financial Economics*, 39(2-3):181–208.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.

Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Working Paper*.

Peng, L. and Xiong, W. (2007). Investor attention and time-varying comovements. *European Financial Management*, 13(3):394–422.

Peyer, U. and Vermaelen, T. (2009). The nature and persistence of buyback anomalies. *Review of Financial Studies*, 22(4):1693–1745.

Tetlock, P. C. (2010). Does public financial news resolve asymmetric information? *Review of Financial Studies*, 23(9):3520–3557.

Vermaelen, T. (1981). Common stock repurchases and market signalling: An empirical study. *Journal of Financial Economics*, 9(2):139–183.